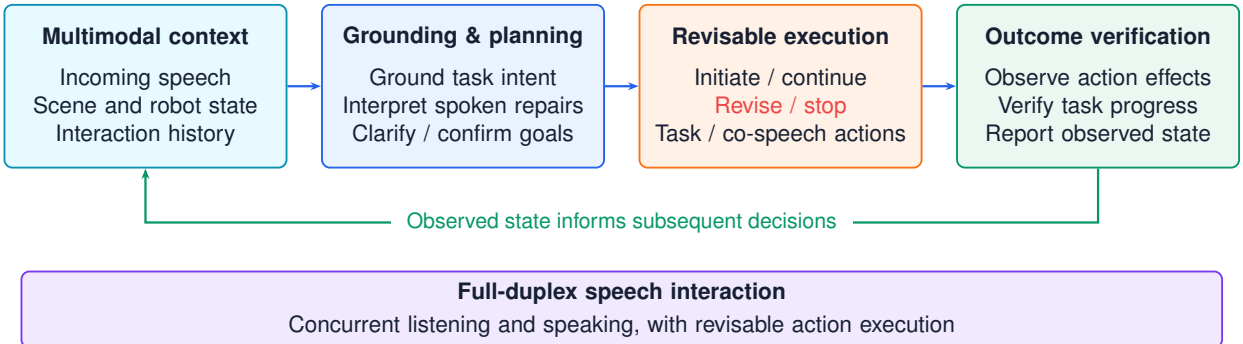
Multimodal context
Incoming speech
Scene and robot state
Interaction history
Grounding & planning
Ground task intent
Interpret spoken repairs
Clarify / confirm goals
Revisable execution
Initiate / continue
Revise / stop
Task / co-speech actions
Outcome verification
Observe action effects
Verify task progress
Report observed state
Observed state informs subsequent decisions
Full-duplex speech interaction
Concurrent listening and speaking, with revisable action execution